

Polar Coordinates: Concepts, Conversion & Graph Sketching

Student: Albert Lee

Date: 2026-05-24

Standard: CCSS.MATH.CONTENT.HSF.TF.??Trigonometric Functions in Polar Form. Convert between rectangular and polar coordinates; sketch polar equations using the Cartesian-trick.

Watch out for these (today's weak spots)

1. **Negative r .** A negative r means the point is on the *opposite* side of the pole.
2. **Angle unit = radians** by default (calculus convention).
3. **arctan quadrant trap.** $\tan \theta = y/x$ alone does NOT pin down the quadrant. Always check the signs of x and y separately.
4. **Choosing r .** Take $r = \sqrt{x^2 + y^2} > 0$ unless the problem allows $r < 0$.
5. **Multiplying by r .** When converting polar $\rightarrow$ rectangular, multiplying both sides by r introduces the pole (origin) as an extra solution ??verify it satisfies the original equation.

Part A ??Why & Formulas

Directions: Show your reasoning. These check that the big idea is solid before you compute.

NOTE ㄹ CONCEPT

Polar coordinates describe a point using **one angle θ** and **one distance r** ??natural for circular motion, awkward for grid shapes. Conversion formulas:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

Trap: $\tan \theta = y/x$ loses the quadrant. Always check signs of x and y .

Worked Example ?? Derive $\tan \theta = y/x$

Start. From $x = r \cos \theta$ and $y = r \sin \theta$ (the two conversion formulas):

Step 1. Divide y by x :

$$\frac{y}{x} = \frac{r \sin \theta}{r \cos \theta}$$

Step 2. The r cancels (assuming $r \neq 0$), leaving

$$\frac{y}{x} = \frac{\sin \theta}{\cos \theta} = \tan \theta. \quad \blacksquare$$

A1. Explain in 2–3 sentences why polar coordinates are *inefficient* for describing the four corners of a square but *ideal* for describing points on a circle.

A2. Given $x = r \cos \theta$ and $y = r \sin \theta$, derive **both** r^2 and $\tan \theta$ in terms of x and y . Show every step (similar style to the worked example above).

A3. Mark each statement **T** or **F** and *justify in one short sentence*.

- a. ☐ T / ☐ F (r, θ) and $(r, \theta + 2\pi)$ are the same point.
- b. ☐ T / ☐ F $(-r, \theta)$ and $(r, \theta - \pi)$ are the same point.
- c. ☐ T / ☐ F $(0, \theta)$ is the pole for *any* value of θ .
- d. ☐ T / ☐ F (r, θ) and $(-r, -\theta)$ are the same point.

Part B ??Plotting & Alternative Representations

Directions: Use a polar grid (sketch in the work space). Read the signs of r and θ both carefully.

NOTE ㉠ MULTIPLE LABELS

A polar point has **infinitely many labels**:

$$(r, \theta) = (-r, \theta + \pi) = (r, \theta + 2\pi k), \quad k \in \mathbb{Z}.$$

Negative r reflects the point across the pole (origin).

Worked Example ??Plot $(-2, 5\pi/6)$ and find two alternative reps

Plot. Rotate to $\theta = 5\pi/6$ (a ray pointing into Q-II at 150°). Since $r = -2$, the point lies on the *opposite* ray, i.e. in Q-IV at $\theta = 5\pi/6 - \pi = -\pi/6$, distance 2 from the pole.

Rep #1 ($r > 0$). Use $(-r, \theta + \pi)$ rule: $(2, 5\pi/6 + \pi) = (2, 11\pi/6)$.

Rep #2 ($r < 0$, different θ). Add 2π to the angle: $(-2, 5\pi/6 + 2\pi) = (-2, 17\pi/6)$.

B1. Plot $\left(3, -\frac{7\pi}{4}\right)$ and state which **quadrant** it lies in. (Watch the negative angle.)

B2. Plot $\left(-4, \frac{13\pi}{6}\right)$. Then give the equivalent rep with $r > 0$ and $0 \leq \theta < 2\pi$.

B3. Given $\left(5, \frac{7\pi}{6}\right)$, find an equivalent representation with $r < 0$ AND $-2\pi < \theta < 0$.

B4. Plot the following four points on **one** polar grid:
 $(2, \pi/3)$, $(-2, \pi/3)$, $(2, -\pi/3)$, $(-2, -\pi/3)$. Which *two* coincide?

Part C ??Point Conversion

Directions: Reduce any over-rotated angles first. For rect $\rightarrow$ polar, give $r > 0$ and $0 \leq \theta < 2\pi$.
Quadrant check is required.

NOTE 𐄞 DIRECTION MATTERS

Polar $\rightarrow$ Rectangular: $x = r \cos \theta$, $y = r \sin \theta$. Just substitute.

Rectangular $\rightarrow$ Polar: $r = \sqrt{x^2 + y^2}$, $\theta = \text{atan2}(y, x)$. Always check the quadrant from the signs of x and y before reporting θ ??naive $\arctan(y/x)$ loses Q-II and Q-III.

Worked Example ??Convert $(-3, 3\sqrt{3})$ rect $\rightarrow$ polar ($r > 0$, $0 \leq \theta < 2\pi$)

Quadrant. $x = -3 < 0$, $y = 3\sqrt{3} > 0 \Rightarrow$ the point lies in Q-II.

Radius. $r = \sqrt{(-3)^2 + (3\sqrt{3})^2} = \sqrt{9 + 27} = \sqrt{36} = 6$.

Reference angle. $\tan \theta_{ref} = \left| \frac{y}{x} \right| = \frac{3\sqrt{3}}{3} = \sqrt{3} \Rightarrow \theta_{ref} = \frac{\pi}{3}$.

Quadrant adjust (Q-II). $\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$.

Answer. $(r, \theta) = (6, \frac{2\pi}{3})$.

C1. Polar $\rightarrow$ Rectangular: $\left(-4, \frac{23\pi}{4}\right)$. *Hint: reduce θ modulo 2π first.*

C2. Polar $\rightarrow$ Rectangular: $\left(6, \frac{11\pi}{6}\right)$.

C3. Rectangular $\rightarrow$ Polar: $(-\sqrt{6}, -\sqrt{2})$. (Tricky — Q-III. Exact form, $r > 0$, $0 \leq \theta < 2\pi$.)

C4. Rectangular $\rightarrow$ Polar: $(0, -5)$. (Edge case — on the negative y -axis. State r and θ directly without dividing by zero.)

Part D ??Sketching $r = a \cos \theta$ / $r = a \sin \theta$

Directions: Use the Cartesian-trick. Sketch the sinusoid first; then read its height as r and convert to a polar point.

NOTE ㉠ THE CARTESIAN-TRICK

Step 1. First sketch the Cartesian $y = a \cos x$ (or $y = a \sin x$).

Step 2. Treat the input x as the polar angle θ . The output y becomes the polar radius r for that angle.

Step 3. When $y < 0$ (i.e. $r < 0$), the polar point lies on the *opposite* ray. Don't worry — the curve still ends up on the same circle.

Result. $r = a \cos \theta$ is a circle of diameter $|a|$ sitting on the x -axis. $r = a \sin \theta$ is a circle of diameter $|a|$ sitting on the y -axis. Sign of a flips the side.

Worked Example ??Sketch $r = -2 \cos \theta$

Step 1. Sketch $y = -2 \cos x$ on Cartesian axes. (A flipped cosine: max $+2$ at $x = \pi$; min -2 at $x = 0$; zeros at $x = \pi/2, 3\pi/2$.)

Step 2. Read off values:

- At $\theta = 0$: $r = -2 \rightarrow$ opposite side, distance 2 from pole $\rightarrow$ point $(-2, 0)$ in rect.
- At $\theta = \pi/2$: $r = 0 \rightarrow$ the pole.
- At $\theta = \pi$: $r = 2 \rightarrow$ angle 180° , distance 2 $\rightarrow$ point $(-2, 0)$ again.
- At $\theta = 3\pi/2$: $r = 0 \rightarrow$ pole.

Step 3. Connect smoothly $\rightarrow$ circle of diameter 2 lying entirely on the **negative** x -axis (center $(-1, 0)$, radius 1). The whole circle is traced over $\theta \in [0, \pi]$; $\theta \in [\pi, 2\pi]$ retraces it.

D1. Sketch $r = 3 \sin \theta$ over $0 \leq \theta \leq 2\pi$. Use the two-panel format.

Step 1: Cartesian $y = 3 \sin x$

Step 2: Polar $r = 3 \sin \theta$

D2. Sketch $r = -4 \cos \theta$ over $0 \leq \theta \leq 2\pi$. Two-panel format. Confirm: which side of the x -axis is the circle on, and what is its diameter?

Step 1: Cartesian $y = -4 \cos x$

Step 2: Polar $r = -4 \cos \theta$

Answer: diameter = _____, circle is on the _____ side of x -axis.

Part E ??Equation Conversion

Directions: Show the substitution step before simplifying. State the final form clearly.

NOTE ⚡ SUBSTITUTION RULES

Rect $\rightarrow$ Polar: sub $x = r \cos \theta$ and $y = r \sin \theta$, then simplify.

Polar $\rightarrow$ Rect: often it helps to multiply both sides by r ; then use

$r^2 = x^2 + y^2$, $r \cos \theta = x$, $r \sin \theta = y$. Verify that the pole (origin) is consistent if you multiplied by r .

Worked Example ??Convert $y = x^2$ (rect) to polar

Sub. $r \sin \theta = (r \cos \theta)^2 = r^2 \cos^2 \theta$.

Divide by r (assuming $r \neq 0$): $\sin \theta = r \cos^2 \theta$.

Solve for r : $r = \frac{\sin \theta}{\cos^2 \theta} = \sec \theta \tan \theta$. The pole ($r = 0$) corresponds to $\theta = 0$ (since $\sin 0 = 0$), which is the origin — on the parabola. Consistent.

E1. Rect $\rightarrow$ Polar: $x^2 + y^2 = 9$. Give the answer with $r > 0$.

E2. Rect $\rightarrow$ Polar: $x^2 - y^2 = 1$. (A hyperbola — use a double-angle identity to compact the answer.)

E3. Polar $\rightarrow$ Rect: $r = \frac{1}{\sin \theta - \cos \theta}$. *Hint:* multiply both sides by $(\sin \theta - \cos \theta)$, then by r if useful.

E4. Polar $\rightarrow$ Rect: $r^2 = \tan \theta$. *Hint:* write $\tan \theta = \sin \theta / \cos \theta$, then multiply both sides by $r \cos \theta$ to introduce x and y . Verify the pole.